

ASSIGNMENT #2

ECBM E4070/E6070, Professor Aurel A. Lazar

Deadline: 11:59PM, Friday, February 20th, 2026

1 Task #1:

Implement the reduced photoreceptor model with dynamics governed by the following differential equations

$$\begin{aligned}\frac{dx_1}{dt} &= \alpha_1 \cdot \lambda \cdot (1 - x_1) - \beta_1 \cdot x_1, \\ \frac{dx_2}{dt} &= \alpha_2 \cdot x_1 \cdot (1 - x_2) - \beta_2 \cdot x_2 - \kappa_2 \cdot x_2 \cdot x_3, \\ \frac{dx_3}{dt} &= -\alpha_3 \cdot x_3 + \beta_3 \cdot x_2, \\ I &= \frac{x_2^n}{x_2^n + c^n} g_{TRP} (E_{TRP} - V),\end{aligned}$$

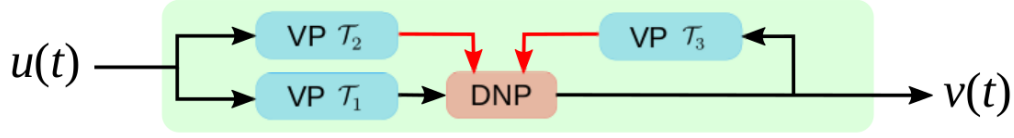
where g_{TRP} and E_{TRP} are respectively the reversal potential of a Transient Receptor Potential (TRP) channel. λ is the light intensity and I is the generated current. Note that the conductance value in the expression of I is modeled via a divisive normalization. Assume that $\alpha_1 = 0.0001$, $\beta_1 = 279.6$, $\alpha_2 = 150.5$, $\beta_2 = 819.8$, $\kappa_2 = 30.0$, $\alpha_3 = 819.8$, $\beta_3 = 4.51$, $c = 0.51$. Set $n = 2$, $g_{TRP} = 1500$, $E_{TRP} = 0$ and $dt = 10^{-5}s$.

The voltage V of the photoreceptor photo-insensitive cell membrane is described by a modified *non-spiking Hodgkin-Huxley* neuron (HHN) model (implementation provided in the **.ipy nb** notebook). For the HHN the current I is the input and the voltage V is the output. Assume that a background light intensity ($\lambda = 1e6$) is applied on the photoreceptor that is followed by a one second step pulse of **different** intensity. Find the steady-state response $V = V(\lambda)$ and plot the light intensity (on a *log* scale) versus steady state membrane voltage. Comment on the shape of the curve.

You will need to use **Euler Method** to simulate the dynamics (example provided in *python_requirement* on Courseworks). Please *read the **comments and doc strings** for non-spiking Hodgkin-Huxley neuron model carefully* before you start.

2 Task #2:

Consider the following feedforward/feedback divisive normalization circuit:



- Part 1. For the transformations $\mathcal{T}^1$ and $\mathcal{T}^2$ the zero order-filters are, respectively, b_1 and b_2 , and the first and second order kernels are, respectively, given by:

$$h_1^1(t) = a_1 \cdot \frac{t}{0.1} \exp(-\frac{t}{0.1}), \quad h_1^2(t) = c_1 \cdot \frac{t}{0.2} \exp(-\frac{t}{0.2}),$$

$$h_2^1(t_1, t_2) = a_2 \cdot 0.0001 \cdot h_1^1(t_1) \cdot h_1^1(t_2), \quad h_2^2(t_1, t_2) = c_2 \cdot 0.0001 \cdot h_1^2(t_1) \cdot h_1^2(t_2),$$

where a_1, a_2, c_1, c_2 are constant parameters. Also the kernels for $\mathcal{T}^3$ are zero (no feedback). Use a wide range of constant light intensities I (i.e. $10, 10^2, \dots, 10^8$) as input $u(t)$ to drive the circuit. Find and plot the steady-state sigmoidal response v as a function of $\log I$. For each of the four parameters a_1, a_2, c_1, c_2 , plot the sigmoidal response as you vary the parameters (choose at least 3-5 values per parameter), label your curves with legends.

- Part 2. Analytically derive (by hand) the steady-state response $v_\infty(t)$ for a constant input

$$u(t) \equiv u_0 \quad (\text{constant}).$$

Assume a purely feedforward (only $\mathcal{T}^1$ and $\mathcal{T}^2$; $\mathcal{T}^3 = 0$) circuit. Show your work in *markdown* format in your notebook (or upload a photo of your handwriting to the submission folder onto *Courseworks*).

Hints:

- Compute the steady-state gains

$$H_1^k := \int_0^\infty h_1^k(t) dt, \quad H_2^k := \int_0^\infty \int_0^\infty h_2^k(t_1, t_2) dt_1 dt_2, \quad k \in \{1, 2\},$$

and use them to obtain closed-form expressions for the steady-state outputs

$$\mathcal{T}_\infty^k(u_0) = b_k + H_1^k u_0 + H_2^k u_0^2, \quad k \in \{1, 2\}.$$

- (Optional) Verify in code that your analytical expression yields the same result as the steady-state computation from *Task 2 Part 1* for several values of u_0 .

Part 3. Analyze the effect of a feedback loop (on the circuit above) described by the transformation $\mathcal{T}^3$ with zero order-filter b_3 and the first and second order kernels:

$$h_1^3(t) = d_1 \cdot \frac{t}{0.04} \exp(-\frac{t}{0.04}), \quad h_2^3(t_1, t_2) = d_2 \cdot 0.0001 \cdot h_1^3(t_1) \cdot h_1^3(t_2),$$

where d_1, d_2 are constant parameters. As above, find the steady-state sigmoidal response v for $u = \log I$, where I is the constant light intensity. Plot the shape of the sigmoid as a function of parameters d_1, d_2 .

3 Submission:

You are provided with a single *homework2.ipynb* as a skeleton for this homework. Helper functions are provided. Note that you do not need to use all the code in the provided code skeleton in your implementation, but you are asked to keep your work in this notebook. Make use of comments and mark-down to keep your code organized. Please submit this notebook with both team members' name included at the top.